APPENDIX I. SIERRA LEONE

JOINT WORLD BANK/IMF DEBT SUSTAINABILITY ANALYSIS BASED ON LOW-INCOME COUNTRY FRAMEWORK—AN UPDATE

1. **This appendix updates the debt sustainability analysis (DSA) presented in March 2006 (IMF Country Report No. 06/183, Appendix IV).** This Low Income Country (LIC)-DSA update, jointly conducted by the Fund and World Bank staffs, is based on (i) the reconciled external debt database and debt service projections prepared for the HIPC completion point DSA, and (ii) revised macroeconomic framework under the PRGF-supported program. Both the debt database and the macroeconomic projections are consistent with those used for the HIPC completion point DSA. The full effect of the MDRI relief, which was shown as a separate scenario in IMF Country Report No. 06/183, is now included in the baseline scenario. The update of the fiscal debt sustainability also takes into account the impact of the gradual conversion of non-interest bearing government securities held by the Bank of Sierra Leone (BSL) into interest bearing securities as well as the government's commitment to provide the BSL with additional securities for monetary operations and recapitalization purposes.[29] The impact of this conversion leads to higher baseline NPV of debt indicators during 2020-26 compared to the debt indicators of the same period reported in the baseline of the previous LIC DSA (March 2006), which did not incorporate the conversion of noninterest bearing liabilities into securities. In the previous LIC DSA, this impact was only analyzed in a separate bound test, which projected significantly higher NPV of debt indicators than in the updated LIC DSA presented here.

2. **The update confirms that Sierra Leone remains at moderate risk of debt distress under the LIC-DSA framework.** A full delivery of HIPC completion point assistance, bilateral assistance beyond HIPC Initiative, and MDRI relief will contribute to bringing debt indicators below the country-specific policy dependent thresholds after 2006.[31] However, stress tests revealed that the Sierra Leone's external debt sustainability is still vulnerable to

[29] The securities will be subject to rollover on maturity.

[31] Sierra Leone remains rated as a poor performer with regard to its policies and institutions under the joint IMF-WB DSA framework for LICs. As a poor performer, the thresholds applied to Sierra Leone are: (i) 100 percent for Net Present Value (NPV) of debt-to-exports, (ii) 30 percent for NPV of debt-to-GDP, and (iii) 200 percent for NPV of debt-to-revenue. The relevant debt service thresholds are (i) 15 percent of exports, and (ii) 25 percent of revenues. ("*Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations*," www.imf.org).

external shocks that are considered under the standard template of the LIC-DSA. Debt burden indicators in some stress tests rise rapidly to above the indicative thresholds especially under the lower export stress test and less concessional borrowing scenario.

Background and Macroeconomic Framework Projections

3. **Sierra Leone's nominal public external debt including arrears stood at US$1,573.2 million (129.6 percent of GDP) as at end-2005.** The largest creditor is IDA (35.4 percent of total debt outstanding), followed by the African Development Fund, and the IMF. Bilateral creditors and commercial creditors accounted for 23.5 percent and 9.2 percent respectively. During the data reconciliation process for the HIPC completion point DSA, the staffs confirmed that the amount of debt to commercial creditors that could be verified by proper documentations is US$144 millions. While the amount could be increased as the verification process for IDA debt buyback progresses, this update used the currently verifiable amount for the analysis to share a consistent database with the HIPC completion point DSA.[32]

4. **The baseline scenario, which is consistent with the HIPC completion point DSA, is built on continuation of prudent macroeconomic management.** The key assumptions include: (i) an average real GDP growth rate at 6¾ percent between 2006 and 2011, then gradually declining to 5.2 percent by 2026, (ii) an increase in export growth in current dollar terms from about 15 percent in the medium term to 5¾ percent in the long term, and (iii) a gradual increase in import growth in current dollar terms from about 9½ percent in the medium term to 6 percent in the long term. The higher output and export growth in the medium-term is based on the continued expansion of rutile and bauxite production and continued strong growth of diamond export. In addition to the mining sector, Sierra Leone's economic growth will be driven by agriculture (rice and cash crops), which is supported by a food security plan as a key policy pillar. With a prudent fiscal policy aimed at fiscal consolidation to allocate enough financial resources for private sector development, the average domestic primary balance will be -2.2 percent of GDP through 2026 compared to -3.1 percent in 2005.

5. **External donor assistance (both grants and loans) is assumed to decrease gradually in terms of GDP over the projection period.** External borrowing is expected to decline gradually from 3 percent of GDP on average in the medium-term to 2 percent by 2026. IDA borrowing is assumed to constitute about 60 percent of new borrowing until 2015, with increased borrowing from other creditors including bilaterals on less concessional terms

[32] The amount could possibly increase by about US$100 million at maximum including to some creditors that have recently not been reachable by the authorities.

during the remaining projection period. External grant is also assumed to decline from over 8 percent of GDP in 2006 to 4 percent by 2026.

External Debt Sustainability

6. **Under the staff baseline scenario, the NPV of the debt indicators will remain below the corresponding thresholds once the completion point of the HICP initiative is reached and the HIPC/MDRI relief is delivered.** The NPV of debt-to-export ratio is expected to decline to 36 percent in 2006 once the country reaches the HIPC completion point and qualify for the MDRI. This scenario reflects the continuous rehabilitation and development of the rich mineral resources that has substantially increased exports in percent of GDP. The NPV of debt-to-GDP ratio would remain substantially below the threshold (30 percent) during the entire projection period. In addition, given the significant impacts of the HIPC and MDRI debt service saving, the debt service ratio would stay around 1-2 percent of exports under the baseline scenario in the medium term, against the 15 percent threshold.

7. **Careful interpretation should be given to the bound tests, which reveal that external debt indicators could deteriorate substantially under the historical scenario and in the face of adverse external development.** Either or both of the NPV of debt-to-GDP and debt-to-exports ratio will deteriorate significantly to levels that are above the corresponding thresholds and would not return to sustainable paths under these two tests (A.1 and B.2 in Table 2). Given that the historical averages and their standard deviations reflect the period of intensified civil conflict and the costly process that followed, these tests indicate the importance of keeping the security that has been achieved.

8. **The test, however, clearly indicate the need for prudent external debt policy.** Under the alternative scenario that assumes external borrowing on less concessional terms (A.2), the debt burden becomes heavier over the long-term and surpass corresponding thresholds without returning to the sustainable path. For example, the NPV of debt-to-exports ratio would reach 105 percent by 2026, and the NPV of debt-to-GDP would approach to its threshold. Therefore, it is a crucial element of prudent debt management to rely on grants or concessional loans to maintain the external debt sustainable.

Fiscal Debt Sustainability

9. **Sierra Leone's large domestic debt represents a significant additional burden.** Adding the NPV of domestic debt, including arrears, to the NPV of external obligations raises the debt-to-GDP ratio in 2005 by 28 percentage points of GDP to a total of 64 percent of GDP (Table 3). The NPV of debt-to-revenue ratio for 2005 stands at 320 percent (if the revenue base includes grants) while the debt-service-to-revenue ratio is 25 percent.

10. **The NPV of debt indicators would fall rapidly after the completion point but increase again by 2020.** Table 4 shows that the NPV of debt-to-GDP ratio drops to 29 percent by 2007 and an average of 25 percent during 2008-19 before increasing to an average of 30 percent from 2020 onward driven by the impact of the conversion of noninterest bearing government securities. The NPV of debt-to-revenue ratio (including grants) follows a similar trend, falling to 134 percent by 2007 and further to an average of 127 percent during 2008-19 followed by an increase to an average of 161 percent from 2020 onward. The debt-service-to-revenue ratio (including grants) increases gradually to 23 percent in 2026 from 12 percent in 2007. All debt sustainability indicators for 2025 are significantly lower than in the comparable bound test (Bound test C1) of the March 2006 LIC DSA. The previous LIC DSA did not include the impact of debt stock relief under MDRI, and it assumed a one-off conversion in 2009 of all noninterest bearing government securities into marketable securities[33]. The current LIC DSA assumes MDRI and a more gradual conversion of government debt into securities, consistent with the need to recapitalize the BSL and provide securities for open market operations.

11. **A lower primary deficit would help reduce debt but hinder PRSP implementation.** The primary domestic deficit in 2005 was 3.1 percent of GDP, while a drop to 1.5 percent of GDP is projected in 2006. The baseline scenario assumes a primary domestic deficit of 2.7 percent of GDP in 2007, which would slowly rise to an average of 3.5 percent during 2008-12 before declining to 2.3 percent of GDP by 2019 where it would broadly remain through 2026. The higher deficit during 2008-12 reflects the initial cost of poverty reduction strategy implementation in a post-conflict country. By contrast, the alternative scenario A2 (Table 4) maintains the primary domestic balance constant at the 2006 GDP share of -1.5 percent. As a result, the NPV of debt-to-GDP ratio falls from 27 percent of GDP in 2007 to 0.3 percent in 2026, compared to 34 percent in the baseline scenario. The tension between high spending requirements and debt sustainability can only be resolved through a combination of concessional borrowing, grant financing, stronger revenue effort, and efficiency gains in the delivery of public services.

12. **Temporary or permanently lower GDP growth represents the most significant risk to fiscal debt sustainability in Sierra Leone.** Reflecting the limited room to adjust expenditures and collect additional revenue, the government would have to recur to significant additional financing in the face of lower GDP growth. If real GDP growth were permanently lower (by 3 percentage points below the baseline rate), then financing with additional borrowing at concessional terms would increase the NPV of debt-to-GDP to 121 percent of GDP by 2026 compared to 34 percent of GDP as in the baseline scenario (Alternative scenario A3 in Table 4). If real GDP growth were temporarily negative (-7 percent

[33] See IMF Country Report No. 06/183, Appendix IV, Paragraph 19.

in 2007 and 2008), then the NPV of debt-to-GDP ratio would increase to 87 percent of GDP by 2026 (Bound test B1 in Table 4).

Conclusions

13. **The staffs conclude that Sierra Leone continues to face a moderate risk of debt distress.** A full delivery of MDRI relief would bring all external debt burden indicators substantially below the corresponding thresholds during the entire projection period. However, some of the standard alternative scenario and bound tests exacerbate debt indicators by breaching or nearing the relevant thresholds, indicating how debt sustainability critically hinges on prudent macroeconomic and external debt management policies.

14. **To ensure future debt sustainability a gradual reduction of the domestic debt stock is as vital as the successful fiscal adjustment programmed under the baseline.** A lower domestic debt stock would lessen the liquidity and the rollover risks associated with the short maturities of this debt. In order to extend the maturity of domestic borrowing, the government should promote the development of the domestic debt market, which would also facilitate the conversion of the current non-interest bearing, non-maturing domestic debt into a longer-term security. Under the current terms at which the government plans to convert this liability (mainly through high yielding treasury bills), debt service is likely to rise substantially and lead to additional borrowing. In light of this, and also to avoid crowding out of the private sector, fiscal tightening or new external borrowing could become necessary. These considerations would warrant a reassessment of Sierra Leone's risk of debt distress at a later stage.

Table 1. Sierra Leone: External Debt Sustainability Framework, Baseline Scenario, 2005-2026 1/
(In percent of GDP, unless otherwise indicated)

	Actual	Historical Average 6/	Standard Deviation 6/	Projections									
	2005			2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2026	2012-26 Average
External debt (nominal) 1/	**168.7**			**127.5**	**88.4**	**82.1**	**76.0**	**69.7**	**63.9**		**45.7**	**30.4**	
o/w public and publicly guaranteed (PPG)	129.5			95.4	59.4	55.8	52.1	48.4	45.0		34.8	29.7	
Change in external debt	-36.0			-41.3	-39.0	-6.3	-6.2	-6.3	-5.8		-3.0	-0.4	
Identified net debt-creating flows	-19.8			-6.6	-3.7	-2.0	-1.9	-1.7	-1.4		0.2	0.1	
Non-interest current account deficit	**6.8**	**6.5**	**5.2**	**5.0**	**4.9**	**4.4**	**4.1**	**4.0**	**3.9**		**4.3**	**3.9**	4.1
Deficit in balance of goods and services	18.0			13.3	12.6	11.5	10.8	10.5	10.4		10.6	10.7	
Exports	22.1			22.6	24.6	25.9	27.1	28.3	27.9		26.2	24.7	
Imports	40.1			35.9	37.2	37.4	37.9	38.8	38.3		36.8	35.4	
Net current transfers (negative = inflow)	-13.1	-6.1	7.2	-10.3	-9.3	-8.8	-8.4	-8.2	-7.9		-8.0	-8.6	-8.1
Other current account flows (negative = net inflow)	2.0			1.9	1.6	1.8	1.7	1.6	1.5		1.7	1.8	
Net FDI (negative = inflow)	**-3.5**	**-0.8**	**1.2**	**-1.7**	**-1.3**	**-1.4**	**-1.4**	**-1.5**	**-1.6**		**-2.0**	**-2.7**	-2.2
Endogenous debt dynamics 2/	**-23.0**			**-9.8**	**-7.3**	**-5.0**	**-4.6**	**-4.2**	**-3.8**		**-2.2**	**-1.0**	
Contribution from nominal interest rate	0.8			0.4	0.2	0.4	0.4	0.4	0.4		0.5	0.5	
Contribution from real GDP growth	-13.1			-10.3	-7.6	-5.3	-5.0	-4.6	-4.2		-2.6	-1.5	
Contribution from price and exchange rate changes	-10.8			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-16.2**			**-34.7**	**-35.3**	**-4.3**	**-4.3**	**-4.6**	**-4.4**		**-3.1**	**-0.6**	
o/w exceptional financing	-2.7			-2.2	-1.9	-2.0	-1.8	-1.5	-1.2		-0.9	-0.3	
NPV of external debt 4/	75.0			40.2	38.3	36.9	35.4	33.3	31.1		23.7	16.8	
In percent of exports	339.5			178.0	155.8	142.3	130.3	117.8	111.5		90.5	68.0	
NPV of PPG external debt	**35.8**			**8.1**	**9.2**	**10.6**	**11.5**	**12.0**	**12.2**		**12.8**	**16.0**	
In percent of exports	161.9			35.9	37.6	40.8	42.5	42.5	43.7		48.8	64.9	
Debt service-to-exports ratio (in percent)	10.8			7.3	2.1	3.9	4.1	4.8	5.5		5.5	7.7	
PPG debt service-to-exports ratio (in percent)	10.0			5.5	0.8	1.1	1.1	1.4	2.2		2.4	3.8	
Total gross financing need (billions of U.S. dollars)	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.2	
Non-interest current account deficit that stabilizes debt ratio	42.8			46.2	43.9	10.7	10.3	10.3	9.7		7.3	4.3	
Key macroeconomic assumptions													
Real GDP growth (in percent)	7.3	5.3	12.6	7.4	6.5	6.5	6.5	6.6	6.6	6.7	5.8	5.2	5.6
GDP deflator in US dollar terms (change in percent)	5.6	-0.8	9.5	12.9	2.9	1.2	1.1	1.6	1.8	3.6	1.0	0.6	1.0
Effective interest rate (percent) 5/	0.5	0.8	0.5	0.3	0.2	0.4	0.5	0.6	0.7	0.5	1.0	1.8	1.2
Growth of exports of G&S (US dollar terms, in percent)	15.3	6.0	22.8	24.0	19.2	13.8	12.7	12.7	7.1	14.9	5.6	5.7	5.7
Growth of imports of G&S (US dollar terms, in percent)	32.9	16.4	18.1	8.6	13.6	8.4	9.2	10.8	7.2	9.6	5.8	6.0	6.0
Grant element of new public sector borrowing (in percent)	...	...	...	44.7	44.1	46.4	49.0	52.3	52.2	48.1	50.3	41.6	46.9
Memorandym item:													
Nominal GDP (billions of US dollars)	1.2			1.5	1.6	1.7	1.9	2.0	2.2		3.1	5.7	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

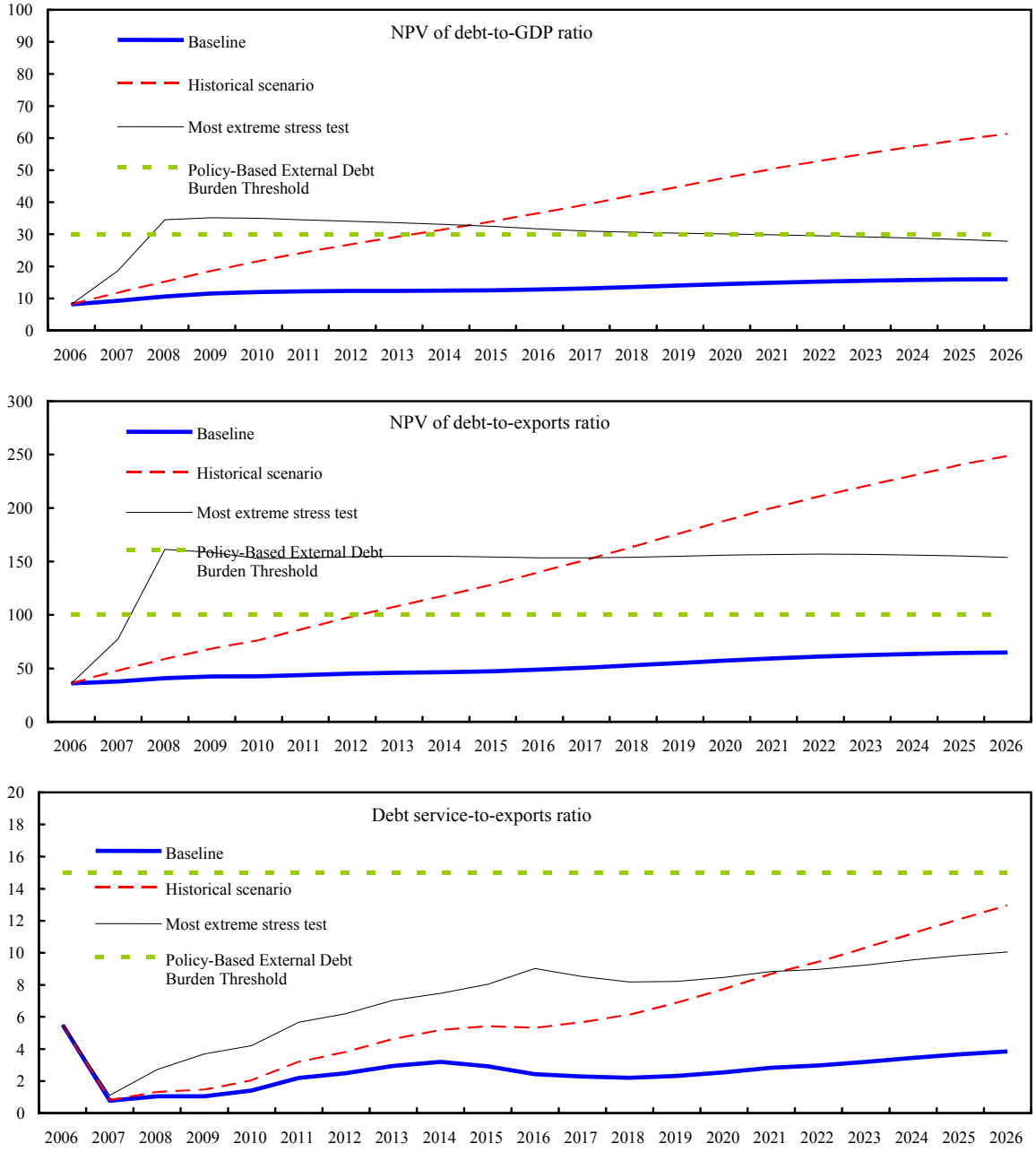
Table 2. Sierra Leone: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006-26
(In percent)

							Projections		
	2006	2007	2008	2009	2010	2011	**2016**	2021	**2026**
NPV of debt-to-GDP ratio									
Baseline	8	9	11	12	12	12	**13**	15	**16**
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2007-26 1/	8	12	15	19	22	24	**37**	50	**61**
A2. New public sector loans on less favorable terms in 2007-26 2/	8	10	12	14	15	15	**19**	23	**26**
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	8	11	14	15	16	16	**17**	20	**21**
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	8	13	21	22	22	22	**20**	20	**19**
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	8	11	14	15	16	16	**17**	19	**21**
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	8	16	23	24	24	23	**22**	21	**20**
B5. Combination of B1-B4 using one-half standard deviation shocks	8	19	35	35	35	34	**32**	30	**28**
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	8	13	15	16	17	17	**18**	21	**23**
NPV of debt-to-exports ratio									
Baseline	36	38	41	42	43	44	**49**	59	**65**
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2007-26 1/	36	48	59	68	76	88	**140**	200	**249**
A2. New public sector loans on less favorable terms in 2007-26 2/	36	40	46	50	52	55	**71**	91	**105**
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	36	38	41	42	43	44	**49**	59	**65**
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	36	77	161	158	153	153	**153**	156	**154**
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	36	38	41	42	43	44	**49**	59	**65**
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	36	64	90	88	84	84	**83**	83	**81**
B5. Combination of B1-B4 using one-half standard deviation shocks	36	82	150	145	139	139	**136**	133	**127**
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	36	38	41	42	43	44	**49**	59	**65**
Debt service ratio									
Baseline	6	1	1	1	1	2	**2**	3	**4**
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2007-26 1/	6	1	1	1	2	3	**5**	9	**13**
A2. New public sector loans on less favorable terms in 2007-26 2/	6	1	1	1	2	2	**3**	5	**6**
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	6	1	1	1	1	2	**2**	3	**4**
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	6	1	3	4	4	6	**9**	9	**10**
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	6	1	1	1	1	2	**2**	3	**4**
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	6	1	2	2	2	3	**5**	5	**5**
B5. Combination of B1-B4 using one-half standard deviation shocks	6	1	2	3	4	5	**8**	8	**9**
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	6	1	1	1	1	2	**2**	3	**4**
Memorandum item:									
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	46	46	46	46	46	46	**46**	46	**46**

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Country: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2006-2026
(In percent)



Source: Staff projections and simulations.

Table 3. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-2026
(In percent of GDP, unless otherwise indicated)

	Actual			Estimate					Projections				
	2005	Historical Average 5/	Standard Deviation 5/	2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2026	2012-26 Average
Public sector debt 1/	199.4			150.5	113.8	101.1	91.9	84.9	78.7		57.2	48.6	
o/w foreign-currency denominated	171.3			127.9	94.2	84.3	77.8	71.3	65.5		46.6	31.0	
Change in public sector debt	-50.1			-48.8	-36.7	-12.8	-9.2	-7.0	-6.2		-3.4	1.0	
Identified debt-creating flows	-38.2			-39.2	-6.5	-9.9	-10.0	-4.7	-3.9		-1.8	1.0	
Primary deficit	-0.1	4.3	2.3	-1.9	0.9	3.9	-0.3	2.9	2.7	1.3	1.4	1.8	1.7
Revenue and grants	20.1			20.0	22.0	20.8	20.2	20.0	20.0		19.3	18.2	
of which: grants	8.3			7.5	34.6	7.4	5.8	5.4	5.1		4.3	3.0	
Primary (noninterest) expenditure	20.0			18.1	22.8	24.7	19.8	22.9	22.6		20.8	20.0	
Automatic debt dynamics	-35.4			-34.7	-5.0	-10.3	-6.2	-5.8	-5.1		-2.3	-0.5	
Contribution from interest rate/growth differential	-19.0			-17.1	-10.3	-7.5	-6.8	-6.0	-5.4		-2.8	-0.9	
of which: contribution from average real interest rate	-2.1			-3.4	-1.1	-0.6	-0.6	-0.4	-0.1		0.6	1.5	
of which: contribution from real GDP growth	-16.9			-13.6	-9.2	-7.0	-6.2	-5.7	-5.3		-3.3	-2.4	
Contribution from real exchange rate depreciation	-16.4			-17.6	5.3	-2.8	0.6	0.3	0.3		...	...	
Other identified debt-creating flows	-2.7			-2.5	-2.4	-3.5	-3.4	-1.8	-1.5		-0.9	-0.3	
Privatization receipts (negative)	0.0			-0.3	-0.5	-1.5	-1.6	-0.3	-0.3		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-2.7			-2.2	-1.9	-2.0	-1.8	-1.5	-1.2		-0.9	-0.3	
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-11.9			-9.6	-30.2	-2.9	0.8	-2.3	-2.3		-1.6	0.0	
NPV of public sector debt	64.4			30.8	29.5	27.6	25.9	25.9	25.7		23.7	33.9	
o/w foreign-currency denominated	36.3			8.1	9.9	10.9	11.8	12.3	12.5		13.1	16.4	
o/w external	36.3			8.1	9.9	10.9	11.8	12.3	12.5		13.1	16.4	
NPV of contingent liabilities (not included in public sector debt)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need 2/	16.8			12.0	12.4	15.6	10.6	13.1	13.9		13.7	20.6	
NPV of public sector debt-to-revenue ratio (in percent) 3/	319.7			154.0	134.2	132.8	128.3	129.5	128.8		122.3	186.3	
o/w external	180.2			40.7	44.9	52.2	58.5	61.5	62.7		67.5	89.8	
Debt service-to-revenue ratio (in percent) 3/ 4/	25.2			17.7	11.8	13.5	13.2	14.2	15.5		15.5	22.6	
Primary deficit that stabilizes the debt-to-GDP ratio	50.0			46.9	37.6	16.7	8.8	9.9	8.9		4.8	0.8	
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	7.2	5.3	12.6	7.3	6.5	6.5	6.5	6.6	6.6	6.7	5.8	5.2	5.6
Average nominal interest rate on forex debt (in percent)	0.5	0.9	0.5	0.3	0.2	0.4	0.5	0.6	0.7	0.5	1.0	1.8	1.3
Average real interest rate on domestic currency debt (in percent)	-2.6	-2.6	9.6	-6.3	0.3	2.7	3.6	5.1	6.1	1.9	9.9	9.8	9.2
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.2	2.7	19.5	-11.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	12.9	13.5	10.4	15.7	10.5	7.4	6.5	6.6	6.5	8.9	5.5	4.7	5.3
Growth of real primary spending (deflated by GDP deflator, in percent)	11.4	9.4	27.5	-3.1	34.6	15.1	-14.5	23.3	5.1	10.1	5.0	4.6	4.7
Grant element of new external borrowing (in percent)	0.0	0.0	0.0	44.7	44.1	46.4	49.0	52.3	52.2	48.1	50.3	41.6	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

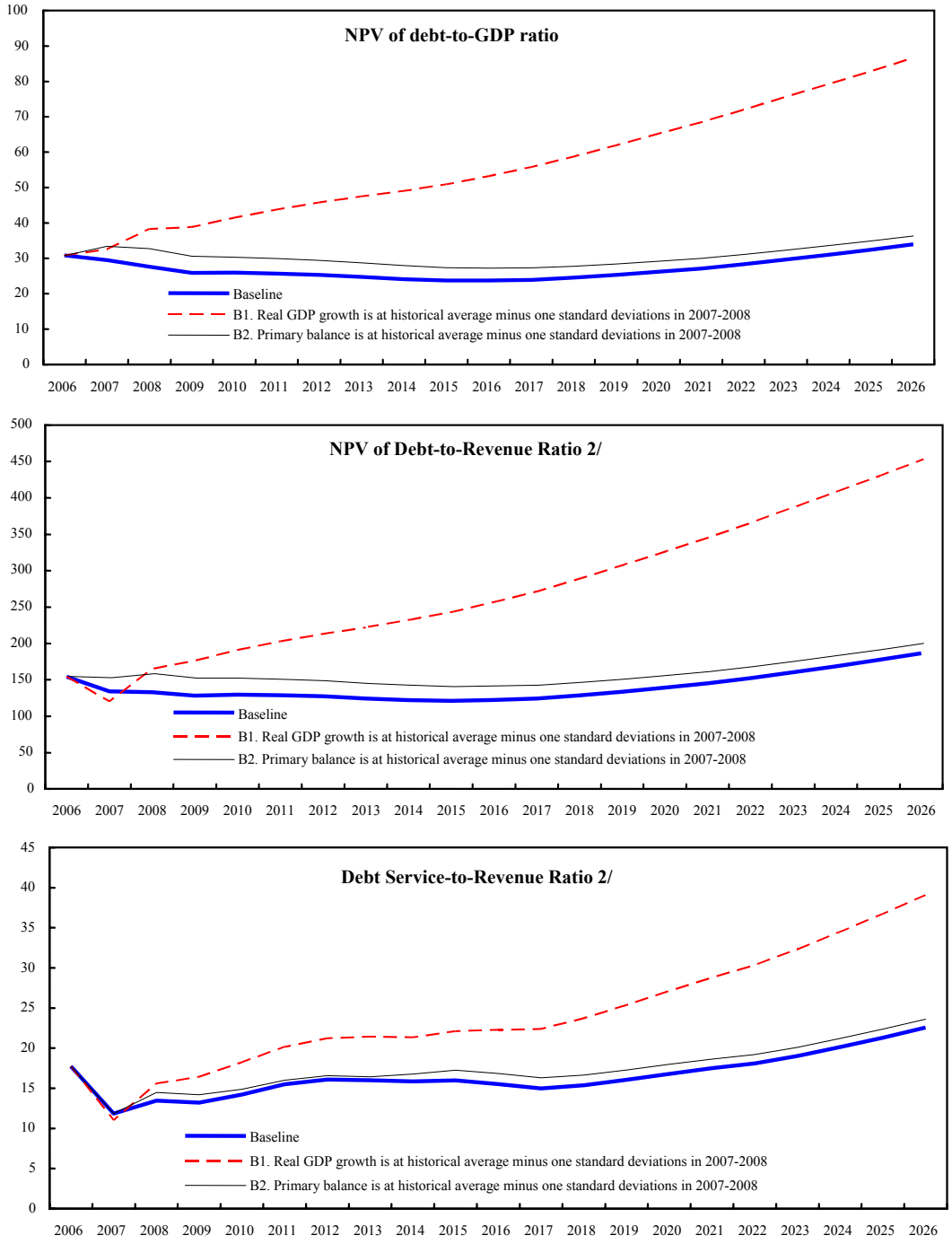
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Sierra Leone: Sensitivity Analysis for Key Indicators of Public Debt 2006-2026

	Projections							
	2006	2007	2008	2009	2010	2011	2016	2026
NPV of Debt-to-GDP Ratio								
Baseline	31	29	28	26	26	26	24	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	31	32	30	32	33	34	39	60
A2. Primary balance is unchanged from 2006	31	27	22	20	17	14	4	0.3
A3. Permanently lower GDP growth 1/	31	30	29	29	31	33	45	121
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008	31	33	38	39	41	44	53	87
B2. Primary balance is at historical average minus one standard deviations in 2007-2008	31	33	33	31	30	30	27	36
B3. Combination of B1-B2 using one half standard deviation shocks	31	35	35	32	31	30	25	33
B4. One-time 30 percent real depreciation in 2007	31	50	47	44	44	43	37	44
B5. 10 percent of GDP increase in other debt-creating flows in 2007	31	36	33	31	31	31	28	37
NPV of Debt-to-Revenue Ratio 2/								
Baseline	154	134	133	128	129	129	122	186
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	154	144	145	157	161	165	198	321
A2. Primary balance is unchanged from 2006	154	125	105	98	84	71	22	2
A3. Permanently lower GDP growth 1/	154	131	139	141	150	159	220	593
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008	154	120	165	176	191	203	257	452
B2. Primary balance is at historical average minus one standard deviations in 2007-2008	154	152	158	151	151	150	141	199
B3. Combination of B1-B2 using one half standard deviation shocks	154	141	161	154	151	147	126	178
B4. One-time 30 percent real depreciation in 2007	154	229	226	221	218	214	190	242
B5. 10 percent of GDP increase in other debt-creating flows in 2007	154	165	160	155	155	153	144	201
Debt Service-to-Revenue Ratio 2/								
Baseline	18	12	13	13	14	15	16	23
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	18	12	14	14	16	17	19	31
A2. Primary balance is unchanged from 2006	18	12	13	12	13	14	12	13
A3. Permanently lower GDP growth 1/	18	12	14	14	16	17	21	47
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008	18	11	16	16	18	20	22	39
B2. Primary balance is at historical average minus one standard deviations in 2007-2008	18	12	14	14	15	16	17	24
B3. Combination of B1-B2 using one half standard deviation shocks	18	11	16	15	16	17	18	24
B4. One-time 30 percent real depreciation in 2007	18	12	14	14	15	17	17	25
B5. 10 percent of GDP increase in other debt-creating flows in 2007	18	12	15	14	15	16	17	24
Debt Service-to-GDP Ratio								
Baseline	4	3	3	3	3	3	3	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4	3	3	3	3	3	4	6
A2. Primary balance is unchanged from 2005	4	3	3	2	3	3	2	2
A3. Permanently lower GDP growth 1/	4	3	3	3	3	4	4	10
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008	4	3	4	4	4	4	5	7
B2. Primary balance is at historical average minus one standard deviations in 2007-2008	4	3	3	3	3	3	3	4
B3. Combination of B1-B2 using one half standard deviation shocks	4	3	3	3	3	4	4	4
B4. One-time 30 percent real depreciation in 2007	4	3	3	3	3	3	3	4
B5. 10 percent of GDP increase in other debt-creating flows in 2007	4	3	3	3	3	3	3	4

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure 2. Sierra Leone: Indicators of Public Debt Under Alternative Scenarios, 2006-2026 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2016.
2/ Revenue including grants.